10 Supporting Consistent Design and Sensemaking across Ship Bridge Equipment through Open Innovation

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INTRODUCTION

A ship's bridge is a complex work environment that is typically outfitted with a range of equipment supplied from numerous different vendors (Lützhöft & Vu, 2018). Multivendor ship bridge systems (MBS) that merge independent equipment from different vendors can create disparities for navigation crew once all necessary equipment is installed into a single work environment (Nordby, Mallam & Lützhöft, 2019). Navigators must engage with different pieces of equipment located on the bridge of a ship throughout their watch shift to enable successful planning and execution of operations. Thus, the very nature of MBS create circumstances where navigators must interact with equipment having multiple design languages across the many physical and digital inputs and outputs of a bridge (see Figure 10.1). For example, different software systems, even supplied by the same vendor, may have different screen layouts, menu structures, colour combinations, iconography, font style and sizes that vary across equipment. A simple analogy to the inconsistencies of typical MBS are the differences found between using contemporary Mac and Windows PC operating systems. Users must adapt to the individual systems and functionalities in order to successfully complete desired tasks. Poor design increases cognitive demands for users (Woods & Patterson, 2000) and can have negative consequences, particularly in the safety-critical context of operations at sea (Lee & Sanquist, 2000; Mallam, Lundh & MacKinnon, 2015). Poor Graphical User Interface (GUI) design has shown to have negative implications on navigation operations, increasing the potential for making errors, hiding critical information and contributing to accidents and deaths at sea (Kataria, Praetorius, Schröder-Hinrichs & Baldauf, 2015; Mallam, Nordby, Johnsen & Bjørneseth, 2020).

This chapter outlines how current conditions in the maritime industry, regulatory scope and ship procurement processes that create an ecosystem where poor design



FIGURE 10.1 Example of an multivendor ship bridge systems (MBS) centre console of a contemporary patrol vessel.

solutions can become more readily introduced and embedded in bridge design, and thus affect seafarers and their sensemaking during operations. The OpenBridge Design Guideline is presented, which addresses these deficiencies by developing interface guidelines based on web technologies, user-centred design principles and component libraries. We detail the background and philosophy of OpenBridge, including how it fills gaps in current industry processes and regulatory frameworks, and strive for a more integrated approach through open innovation. Finally, we present a case study applying the OpenBridge Design Guideline to the development of an ECDIS (Electronic Chart Display and Information System) and discuss its contributions for enhancing sensemaking in ship operations.

CURRENT SHIP PROCUREMENT PROCESSES INTRODUCE INHERENT DESIGN DEFICIENCIES

There are a range of influencing factors for why inconsistent bridge systems are prevalent across the maritime industry. The typical contemporary procurement process for ship bridges have different systems provided by multiple vendors; however, this is not in and of itself the reason for such drastic design inconsistencies across bridge systems. The design and construction of ships are large-scale projects that typically take years from initial concept to a constructed vessel sailing in water (Eyres & Bruce, 2012; Veenstra & Ludema, 2006). Shipbuilding processes are often split between numerous stakeholders and geographical locations (Stopford, 2009; Österman, Ljung & Lützhöft, 2009). Ultimately, ship owners and investors generally focus on big picture issues of a ship's construction and specifications, such as cargo carrying capacity, speed, versatility and efficiency, as opposed to detailed design of the working environment (Eyres & Bruce, 2012).

Typical ship design processes are inherently engineering centric (e.g. ship design spiral [Evans, 1959]) and normally do not include the perspectives and knowledge of the seafarers themselves, or operational demands within design cycles (de Vries, Costa, Hogström & Mallam, 2017). Furthermore, those in charge of maritime equipment design generally do not have an understanding of operations or seafarer demands, leading to poor design choices (Chauvin, Le Bouar & Renault, 2008; United States Fleet Forces Command, 2017). A technology-centric implementation of bridge systems can lead to designs that do not support the users or sensemaking in operations (Johnsen, Kilskar & Danielsen, 2019). Without expert user knowledge integrated throughout design development, a disconnect can emerge between the final ship design, including its onboard equipment, and how the crew use the ship to accomplish their tasks (Mallam, Lundh & MacKinnon, 2015). This can lead to suboptimal and unsafe working practices, increasing the likelihood of errors and accidents.

Current bridge mandatory design regulations and non-mandatory design guidelines also fail to adequately support design consistency across bridge equipment (Mallam & Nordby, 2018). The International Maritime Organization's *International Convention for the Safety of Life at Sea* (SOLAS Convention) provides predominantly goal-based objectives for bridge work environment design (e.g. SOLAS chapter V/15) (International Maritime Organization, 2009). Additional guidance notes and

references are provided in a relatively more prescriptive approach to bridge design (e.g. International Electrotechnical Commission, 2008; International Maritime Organization, 2000, 2004; International Organization for Standardization, 2007). However, in totality, detailed aspects of digital interface design guidance are still lacking. Efforts have been made by regulators and industry stakeholders to standardize aspects of GUI across the industry, such as symbols (International Maritime Organization, 2019a, 2019b) and for new technology development and deployment, such as the introduction of ECDIS (International Maritime Organization, 2017). Even guidelines from supporting classification bodies and regulatory authorities fail to capture necessary design guidance, not only for the bridge but for other operational areas of a ship, including the engine department (Mallam & Lundh, 2013).

As equipment and operations increasingly digitize, there is an opportunity to create design guidelines that support design consistency across all systems for multivendor and integrated bridges. Creating design consistency across equipment in the bridge will lead to positive outcomes in navigational operations and support-enhanced sensemaking of seafarers. This cannot be captured in regulations or design guidance alone. Rather, it must also be integrated with practical and usable processes that enable the application of knowledge and design guidance throughout ship and bridge development. This must engage the identified industry stakeholders and ultimately add value to the process and its outcomes.

SUPPORTING SENSEMAKING THROUGH DESIGN

Sensemaking is the process of turning circumstances into a situational understanding in which we can react upon (Weick, Stutcliffe & Obstfeld, 2005; Taylor & Van Every, 2000). By emphasizing sensemaking in the design process we aim to rationalize the user's actions by understanding how cognitive activities are employed to experience meaning in an interaction situation (Weick, 1993). These activities involve perceiving and interpreting sensory data, formulating and using information, and managing attention (Bartscherer & Coover, 2011). Although complex information spaces, such as a ship's bridge, continue to expand extensively as technology and automation develops, human capabilities and information processing struggle to adapt and successfully manage the complexity of operational systems (Bainbridge, 1983). According to Chia (2000), conception is shaped through "differentiating, fixing, naming, labelling, classifying and relating" sense-impressions systematically to construct our experience of social reality. Therefore, digital sensemaking requires mediation in order to meet the user's needs by labelling and categorizing content and choices in user interfaces to balance the streaming of experience (Weick, Sutcliffe & Obstfeld, 2005). Consistent user interface design is a strategy for digital sensemaking that not only supports easier interpretation, improved usability and productivity but also enhances the possibility to transfer skills across systems because users are able to predict how interfaces will look and function due to familiarity and experience (Nielsen, 2002). This makes the user more "fluent" when operating known systems and even new systems on first encounter. This is particularly relevant for many sectors of the maritime domain due to the nature and variability in how seafarer employment is organized. Although some seafarers may work within a single company, or even a single ship for years, there is a

segment of the population that are employed in a relatively transient manner. Seafarers may work on different types of contracts and sign on/off different ships and ship types (whether within a single company, across multiple companies or across industry sectors). Work periods at sea may be organized by different durations (e.g. days, weeks or months at a time) and across short- or long-term work contracts (e.g. a single trip at sea for a specific time period vs. multi-trip contracts and more permanent/reoccurring employment opportunities). Thus, seafarers moving between different ships require a high degree of adaptability to potentially unfamiliar and inconsistent work environments, operational systems and GUI (in addition to different work colleagues, cultures, policies, procedures, routes, etc.).

Design consistency is critical in supporting sensemaking during operations, as sensemaking requires interpretation and action, enacting a more ordered environment from which further cues can be drawn to create a shared understanding (Kilskar, Danielsen & Johnsen, 2019). The performance of users is directly influenced by design, and poor solutions can introduce negative effects for the user and the system (Dul, Bruder, Buckle, Carayon, Falzon, Marras, Wilson & van der Doelen, 2012). In order to reduce usability problems and increase quality of interactions, design should provide affordances, or cues, that users can recognize, make decisions from and then implement correct actions in order to fulfil a desired task and outcome (Pucillo & Cascini, 2014). This is true of very simple interactions, from correctly interpreting and turning on a faucet for desired water temperature and rate of flow (e.g. "which handle for warm water?", "What angle or rotation for optimal flow?") or opening an unfamiliar door ("Do I push or pull? Will it open automatically… from the center?") to complex safety-critical interfaces and working environments, such as airplane cockpits or central control rooms of power plants. Similarly, in the maritime context of a ship's bridge, the design quality and design consistency across interfaces and interaction devices between navigator and equipment are critical for reducing user errors and increasing productivity. Design should impart relevant information and effectively communicate how to achieve the intended task or tasks to its users. Interaction with an object, interface or system, whether physical or digital, should easily communicate relevant information to the user that is interpretable and enables optimal decision-making for their intended actions.

Good design enables sensemaking in navigators by supporting and guiding expectations of a system, its current operational status and future requirements. This is becoming increasingly relevant with digitized interfaces and the characteristics of information on screens. In particularly, how information is organized in digital menus and sub-menus has implications for how users navigate through a software program and the ease of access to relevant and desired information. As the investigation of a recent United States Navy accident found (*USS John S. McCain collision with Alnic MC in 2017*), the design of digital systems and how information is presented can mislead or inhibit users from correctly interpreting a situation and operational status (National Transportation Safety Board, 2017).

Sensemaking for Current and Future Shipping Operations

Like most industries, throughout the 20th and 21st centuries automation has led to both higher operational productivity, which, in turn, also reduced the number of required

personnel. Ship operations are increasingly automated and centralized, requiring less direct implementation of tasks but increased monitoring and troubleshooting of automated processes. As the proliferation of digitalization of front–end interaction and display devices increases, the maritime domain finds itself at crossroads for future operational systems. Current design guidelines fail to adequately address detailed design aspects of digitized systems and GUI for maritime systems, while the most robust, well-established design guidance focuses on the physical aspects of bridge work environments, console construction and layout (Mallam & Nordby, 2018). This presents a timely opportunity to develop and implement common frameworks for digital GUI systems that are currently lacking within the maritime domain.

There are additional spin-off benefits for consistency of control stations (i.e. the bridge, engine control room, shore-side control centre) for the maritime domain as a whole, including shipping companies, equipment manufacturers, policymakers, seafarers and other relevant stakeholders. For example, classification bodies and regulatory authorities can establish consistency in bridge systems across fleets or industry sectors, creating more streamlined inspection and approval processes. Increased consistency would also allow seafarers to switch between ships while maintaining mastery between interfaces, workstations and work environments. This would have additional benefit of reducing training and familiarization periods between seafarers and equipment. As ships are large monetary investments and have operational lifecycles at sea spanning several decades, retrofitting of a ship structure and its systems are required periodically. Digital systems and GUI are more flexible to develop, cheaper and faster to upgrade, alter and improve in comparison to hardwired and analogue systems (e.g. analogue control console vs. software and screens). Digital systems have a lower threshold for both development and implementation, making updates potentially more economically favourable, as well as more effective in creating better design solutions, not only at the new-build stage but also for retrofitting across a ship's lifecycle.

Furthermore, interest is increasing in unmanned and autonomous vessels that are remotely controlled and monitored. Although still in relatively early phases, MASS (Maritime Autonomous Surface Ships) are developing rapidly in research and development phases throughout the world (International Maritime Organization, 2018; Pribyl & Weigel, 2018; World Maritime University, 2019). Remote control and/or monitoring of unmanned ships at sea are currently being developed through differing shore-side control centre concepts. However, both "traditional" operations of contemporary navigational tasks at sea and future configurations of shore-side control and monitoring will continue to require interactions between people and technology through GUI in some capacity (Kim & Mallam, 2020). The design of equipment and interactions between technology and people must support sensemaking and is perhaps even more critical in the age of distributed command and control of ships at sea where operators and monitors of remote structures will need to have all relevant information to enable optimal sensemaking throughout operations.

TOWARDS IMPROVING DESIGN CONSISTENCY

Consistent design refers to a common set of design principles that are shared by different systems (Nielsen, 2002). Typical examples of such principles are palette definitions or user interface components, such as buttons. Design consistency leads

to a range of benefits, including increased usability, ease of learning and reduced errors (Nielsen, 2002). These benefits increase with the number of systems that apply the same design principles. Although the benefits of consistent design are well known, current maritime design approaches and regulations have not managed to realize the full potential of consistent design. Overall, we argue that each of the individual approaches above have shortcomings and will likely not lead to design consistency alone. Instead, there is a need for a combined approach. Below we describe four design approaches and analyse how they impact the push towards design consistency.

Prescriptive Rules

Prescriptive rules and guidelines specify in detail how a product or system should be designed. This may include detailed design guidance of elements including, font sizes, icons characteristics and specific functions applications must deliver. In the maritime domain, there are both mandatory prescriptive rules and non-mandatory prescriptive guidelines for design. A limitation of prescriptive rules is that they are typically established from prior experiences and thus may have inherent biases in their scope and content. Prescriptive rules related to engineering and technology-related fields are also challenging due to the rapid evolution of contemporary technological advancements. By the time the need for prescriptive rules are identified, developed and released, the applications the rules were originally intended for may have less relevancy or suitability. Currently, there does not exist prescriptive regulation in the maritime sector that defines user interfaces at a level that may lead to digital design consistency. The current rules manage some aspects of central applications, such as ECDIS; however, none offer detailed specifications of user interface design.

Goal-Based Rules

Goal-based rules and guidelines define what a system should achieve, instead of how it should be designed. These guidelines are useful in establishing high-level goals (e.g. SOLAS chapter V/15 [International Maritime Organization, 2009]); however, it is difficult to achieve consistent design using goal-based rules, since a goal can be achieved in many different ways, through many different solutions. Although it is possible to mandate that a system should be designed consistently, it is difficult to realize this in multivendor ecosystems, where many systems are developed independently by independent actors. As a result, a goal-based rule approach for realizing consistent design may work within a single company or group of collaborating companies, but it is likely to fail when systems are designed and integrated by several competing companies.

Manufacturer Controlled Consistency

Typically, larger vendors who control the design of an entire ship's bridge deliver the most consistently designed systems in the maritime sector. These companies may achieve consistency across a single ship's bridge or several ship bridges by controlling

a large portion of the design process and its outcomes. However, this approach is limited to a company's own available systems and product offerings, meaning that ships using equipment from competing manufacturers will no necessarily share the same design principles. Because a single manufacturer rarely delivers all equipment of a single workplace, installed third-party equipment will still create inconsistencies within a bridge work environment.

Design Transfer from Other Fields

In recent years, design principles from the web and mobile industries have begun to enter the maritime domain driven by low-cost, high-quality design guidelines and technologies. These tools are slowly entering the maritime domain across systems in the form of user interface components and design patterns. Over time, this may contribute to system consistency, independently of system vendor. However, there are challenges using such components since their design guidelines were not originally intended for a maritime context and lack specific maritime-related content.

THE OPENBRIDGE CONCEPT

OpenBridge is an initiative that began with the aim to improve the usability of user interfaces in maritime workplaces through collaboration between maritime system developers (Nordby, Frydenberg & Fauske, 2018). The consortium is developing an open innovation approach to maritime workplace design by offering common design guidelines, implementation methods and approval systems. The OpenBridge vision is to simplify both the design and implementation of maritime systems in order to reduce development costs and improve the usability of product offerings.

OpenBridge is based on precedence from the web industry, where freely available frameworks for user interface development have been widely adapted, resulting in reduced costs and improved usability. In order for success, any new design system initiative must be desirable for the commercial interests of its users and the industry at large. Consequently, ensuring the design system is attractive from an economic perspective is of high priority. This includes developing reusable design patterns, tools supporting both design and implementation processes, and reusable code. Reusable code has a dual function: first, it reduces the cost of technical implementation. Second, it reduces the need for developers to interpret the design guidelines when creating new interfaces, since many of the components described in the guideline are already created and catalogued.

The OpenBridge Approach to Consistency

Navigators are constantly interacting with internal bridge equipment (e.g. electronic charts, RADAR, conning display, communication channels), other onboard crew and departments of the ship (e.g. engine, deck crew) and the external environment (e.g. weather conditions, geography, sea traffic, port authorities, vessel traffic services). Individual bridge equipment itself comprises a hierarchy of elements, which in combination create the interaction experience of its user interfaces, including hardware

and software components. Each component of the bridge affects navigators' sensemaking and the totality of their user experience. Thus, the design of equipment can be optimized for each of these components and ultimately extended to MBS integration of other equipment into a single working environment.

The OpenBridge Design Guideline is a component-oriented, prescriptive framework that seeks compatibility with existing prescriptive and goal-based regulations. A core concept is that the OpenBridge Design Guideline is divided into a series of independent components described through a user interface architecture (Nordby, Gernez & Mallam, 2019). The component structure draws upon the design guidelines as a hierarchy of elements that together make up OpenBridge-compatible user interfaces. The user interface architecture includes components, such as palettes, typography and layout patterns. Each element defined in the guideline follows current mandatory regulations to ensure that vendors following the OpenBridge Design Guideline also adhere to industry standards.

When designing a new system, a developer may use every element from the OpenBridge guideline or just a selection of them. In our perspective, designing for consistency should be a long-term and gradual process. It is not a binary output where a system is deemed "consistent" or "not consistent" but rather how many and which consistent attributes a system possesses. In this perspective, we suggest that two different applications are marginally more consistent if they share the same palettes, even though the rest of the applications may be completely inconsistent.

Importantly, OpenBridge focuses on generic user interface components and some core functions requiring consistency. Examples include a standard navigation menu and alarm display that are shared across all bridge systems. However, a degree of freedom is built into the OpenBridge Design Guideline in how to design the core functions of an application. We argue these parts of a system are better served by goal-based guidelines for generally "acceptable" usability levels. Thus, developing OpenBridge is a matter of balancing prescriptive generic user interface design with goal-based functions. Different industry stakeholders will have varying needs and apply the guideline partially or in its entirety (see Table 10.1). For example, integrators and system developers that are dependent on each other have an immediate benefit of fully implementing the OpenBridge Design Guideline. OpenBridge-compatible equipment can be sold to more ships without changes, and integrators can choose between more equipment that is out-of-the-box ready to use without further need for adaptation.

ECDIS CASE STUDY: APPLYING OPENBRIDGE FOR ENHANCED SENSEMAKING

The Design Process

In the following sections, we present how the OpenBridge Design Guideline was used to design a version of an ECDIS user interface. The ECDIS design was developed in an iterative process that included analysis of prior art and in collaboration with domain experts, including developers, equipment manufacturers and maritime

TABLE 10.1
Examples of Different Types of Stakeholders That Have Different Needs in Applying the OpenBridge Design Guideline

Stakeholder Type	OpenBridge Implementation
Independent integrator	They can pick and choose any of the compliance elements
Independent equipment manufacturer not concerned about integration	They can pick and choose any of the compliance elements
Integrator that wants to integrate OpenBridge equipment	Need to follow rules up to integration system structure
Equipment manufacturer that wants to adapt to OpenBridge integrators	Need to follow rules up to applications structure
Integrator that wants to maintain a non-OpenBridge compliant look but integrate third party equipment	Need to follow minimal integration conventions

personnel. Through this initial process we have uncovered several challenges across different areas of the design guideline that provides feedback for further refinement and improvement for its content and application in future design processes. It is important to note that the example presented has not been through an approval process and must be considered as an early-phase development sketch. However, this case study sheds light on how different levels of consistency offered by OpenBridge are applied and how it affects users' sensemaking at each level.

Palettes

OpenBridge has four palette settings for maritime user interfaces: bright, day, dusk and night (see Figure 10.2). The user interface components in OpenBridge are built upon these palettes, meaning that developers of maritime applications do not need to invest resources on colour palette or colour contrast development challenges. The palette system affects sensemaking in three ways. First, it is designed to offer optimized contrast and readability, which supports general sensemaking. For example, the "day" and "bright" palettes offer Level AAA contrast relative to Web Content Accessibility Guidelines (WCAG). Second, the palette uses its colour scheme to create meaning. For example, fonts, numerical data and interface sections are coded with consistent colours to improve the user's ability to decipher the content and meaning within the user interface. This has the potential to improve sensemaking of a single user interface, and when applied to several interfaces, it will also contribute to improving users' sensemaking across interfaces. Third, in ship bridges, different palettes can create light pollution problems for users. A common palette across systems will help reduce these inconsistencies. This has a particular advantage on the bridge during operations in the dark, where small differences in screen light emittance levels can impact the ocular adaption of seafarers on duty more so than in other conditions.

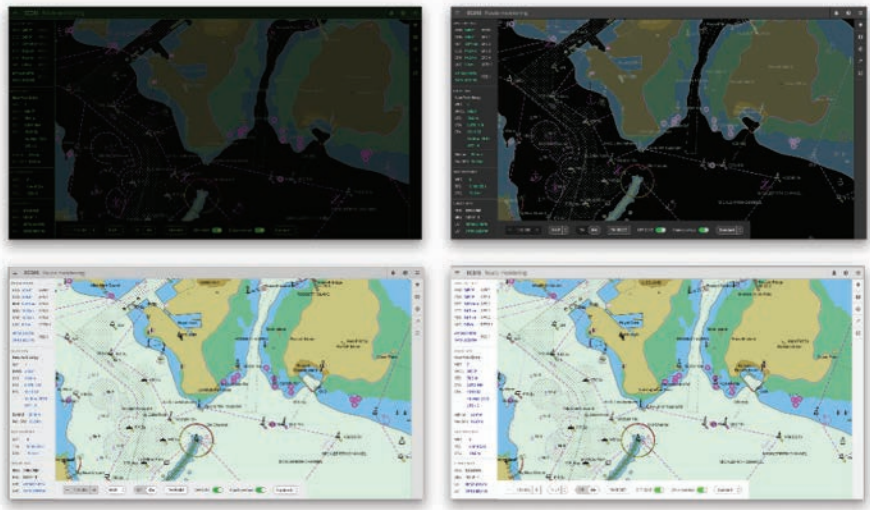


FIGURE 10.2 Examples of electronic chart display and information system (ECDIS) interfaces with different palettes.

Typography

OpenBridge typography was established to have a limited set of text with distinct styles to represent specific types of information. Designers then apply the different text styles according to the specified functions in the interface, thus increasing design development efficiency and consistency by removing the need to create individual typography systems. Typography affects sensemaking in a similar way as palettes. First, OpenBridge has a limited choice of typesets, which requires designers to simplify how typography is used. Second, the typography is coded to help users navigate the interface, generating a clear separation between different elements (e.g. headings, labels and instrument text), thus supporting the users' ability to locate desired information within the interface and reduce screen clutter. Third, the meaning of the different typesets are standardized. Thus, typography provides design affordance, or cues, embedded within the typeset that users will learn, and maintains consistency across all other OpenBridge-compatible interfaces.

User Interface Components

The OpenBridge Design Guideline has a selection of user interface components that can be used to build an interface. This ranges from simple components, such as buttons, to complex nested components, such as the application navigation menu. The design guideline also includes maritime-specific components, such as compass and rudder. The ECDIS user interface was created using ready-made components already established in the guideline. The user interface components support sensemaking in offering predictable, consistent information presentation and interactions. In previous user

interfaces, simple components, such as toggle buttons, could have radically different designs across different equipment (Nordby, Mallam & Lützhöft, 2019). By standardizing a set of common components, it is easier for users to understand key functions. We argue that an additional effect for improved sensemaking is that more predictable user interface components allow users to free their cognitive resources on actual maritime operations rather than in learning and interpreting the operational systems.

Generic Applications Structure

OpenBridge introduces a strict application structure that is governed by a top bar (see Figure 10.3). This structure organizes generic functions (i.e. functions that most applications offer, e.g. navigation menu, alerts, dimming and applications centre) in a predictable manner across systems. In the navigation menu, specific functions have a fixed position. This concept allows users that have learned the OpenBridge structure to immediately know where to find typical functions within the interface.

The ECDIS case was developed in conjunction with RADAR development and was part of an integrated navigation system. In order to navigate the system, we applied a model that separates between operations, applications and tasks. Operations are managed on a workstation level, where a user can select an operation that would control what applications are visible on the stations around them, such as conning,

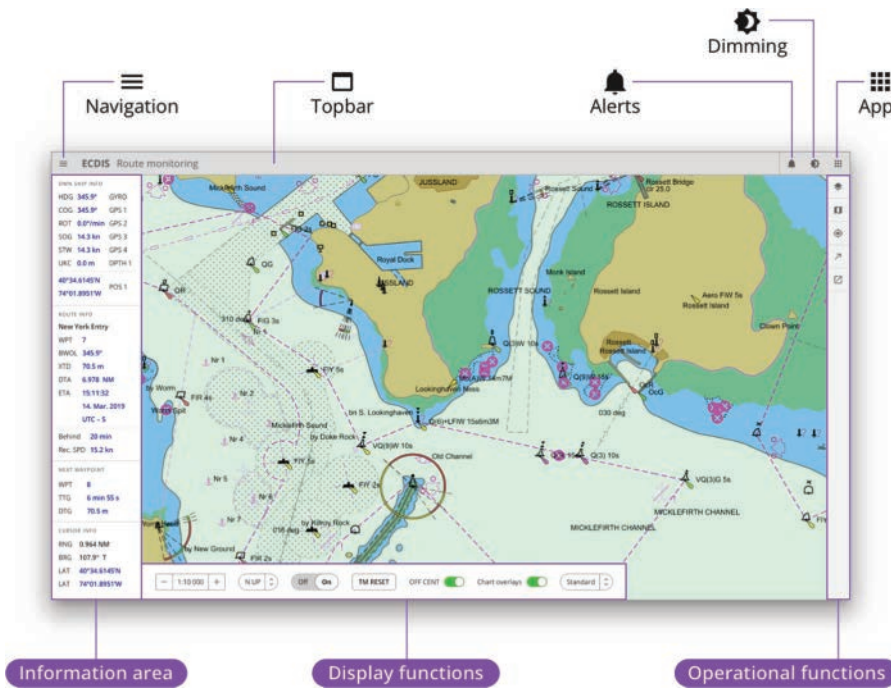


FIGURE 10.3 OpenBridge electronic chart display and information system (ECDIS) interface structure.

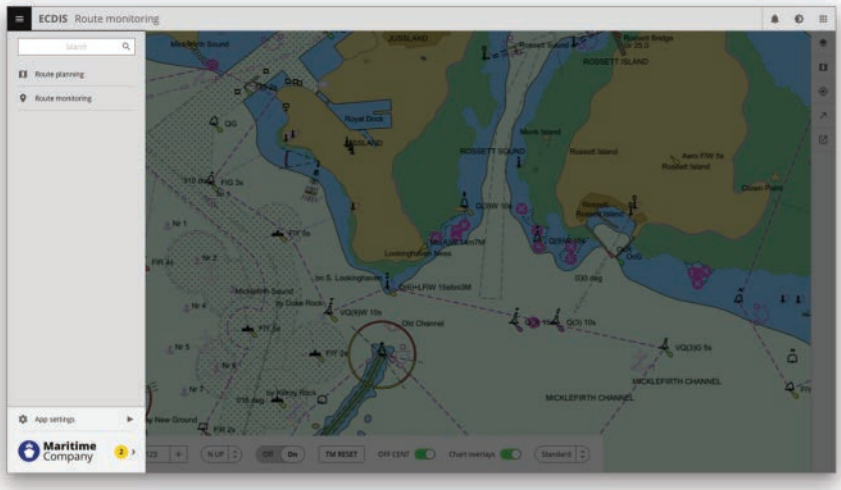


FIGURE 10.4 Application navigation menu (*left*).

ECDIS and RADAR. Each of these applications are managed through tasks, which are accessed through the application navigation menu (see Figure 10.4). The generic applications structure supports sensemaking in helping users to understand where typical applications functions are placed in an application. This is especially important for systems where users need to utilize multiple applications.

Navigation and Information Placement

After having applied the basic elements of the OpenBridge Design Guideline, the main functions of the ECDIS system were then implemented. OpenBridge does not offer any strict guidelines for how this should be performed, but rather establishes the structure around the navigation and information placement to sit within. For ECDIS design, the main functions in the display are separated into interactive functions (right side) and information elements (left side) (see Figure 10.5). The view control is positioned inside the map view to emphasize the connection between the map and the controls.

CONCLUSIONS

The ECDIS case reveals how the various types of elements defined in the OpenBridge Design Guideline supports users' sensemaking in individual applications, but more importantly, by making key attributes consistent with other applications. We also show how many aspects of an interface can be replaced by reusable components that are also shared across other interfaces. This feature has a dual effect: components may achieve improved usability while simultaneously saving development costs for



FIGURE 10.5 Electronic chart display and information system (ECDIS) main functions separated into interactive functions (*right side*) and information elements (*left side*).

manufacturers. This case also reveals how the prescriptive nature of the OpenBridge Design Guideline can be applied in conjunction with goal-based guidelines. The prescriptive guidance of OpenBridge fills current gaps for digital user interface design while complimenting the relatively higher-level general guidance currently in existence within the maritime domain. The OpenBridge Design Guideline is applicable and relevant not only for the maritime domain but can be expanded to other industry's digital interfaces and systems that require consistency. As components of the OpenBridge Design Guideline are generic, it is possible to generalize and apply the fundamental properties to non-maritime uses, expanding the relevancy and broader impact of the guideline for other industries facing similar challenges with digital transformation.

We argue for an open approach for innovation. The strength of an open innovation approach is through momentum and adoption: the value, impact and quality of OpenBridge and the OpenBridge Design Guideline increases as more actors use, test and contribute to its development. The OpenBridge Design Guideline is neither static nor definitive. Rather, it must evolve and change as the industry, technologies and operations evolve and change. As OpenBridge is increasingly applied to real-world applications and design cases the more robust and relevant it will become, ultimately improving design outcomes and sensemaking for end users.

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